

# Monika Battula

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## EXPERIENCE

### Software Engineer Intern — Vertex AI Labs

Jan 2025 – Mar 2025

- Developed scalable backend services using **Java, Spring Boot, and Flask APIs**, improving application throughput by 30%.
- Applied **object-oriented design** and Agile development practices for modular software delivery.
- Integrated automated testing (JUnit, PyTest) and CI/CD pipelines, reducing deployment time by 40%.

### Software Developer Intern — NIT Calicut

May 2024 – Jul 2024

- Built an autonomous robotic arm using **Raspberry Pi, integrating YOLO-based vision and DBSCAN/GMM clustering** for object detection and grasp planning.
- Simulated and validated robotic control, kinematics, and sensor integration using Simulink and Gazebo (RViz) prior to hardware deployment.
- Created front-end dashboards and backend logic using **Node.js, React, and SQL** to visualize real-time telemetry data.
- Refactored C++ computation modules, improving runtime efficiency and reducing memory consumption.
- Collaborated in sprints using JIRA and Git to deliver features aligned with engineering timelines.

## PROJECTS

### Plant Disease Detection As A Service

Oct 2025 – Dec 2025

- Built an end-to-end CNN-based plant disease detection service with image preprocessing, model training, evaluation, and real-time inference exposed via REST APIs for mobile and web clients.
- Containerized and deployed a scalable backend architecture enabling production-ready ML inference.
- Technologies: Python, CNN, TensorFlow/PyTorch, Flask/FastAPI, React Native, Docker, Git

### Legal Document Assistant

Nov 2025 – Dec 2025

- Developed a full-stack AI-driven legal document automation platform using conversational LLMs to guide users through form completion and structured document generation.
- Implemented backend services for session handling and dynamic DOCX processing with scalable API-driven workflows.
- Technologies: Python, FastAPI, React, TypeScript, Tailwind CSS, Google Gemini API, python-docx, SQLite, Git

### AI-Based Multilingual Video Dubbing System

- Built an end-to-end AI dubbing pipeline integrating speech recognition, speaker diarization, translation, and regional text-to-speech for multi-speaker, multilingual video localization.
- Designed a segment-level lip-sync and audio alignment module using duration-aware speed adjustment to ensure realistic dubbing output.
- Technologies: Python, Whisper, Pyannote.audio, Google Cloud TTS, Indic Translator, MoviePy, FastAPI, React, Docker, Git

### Customer Support Email Automation with AI Agents + RAG

- Designed a multi-agent AI system to automatically monitor Gmail, classify customer emails, and generate high-quality responses using LangGraph-based orchestration.
- Technologies: LangGraph, LangChain, FastAPI, LangServe, Groq Llama-3.1-70B, Gemini Embeddings, FAISS/Chroma, Gmail API, Python, Docker

## EDUCATION

### University of Maryland, Baltimore County (UMBC)

Aug 2025 – May 2027

Master of Science in Computer Science (Software Systems & AI) | GPA: 4.0/4.0

Relevant Coursework: Computer Networks, Algorithms, Machine Learning, Cloud Computing

## TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, SQL, Bash
- **Machine Learning & Deep Learning:** CNNs, Transfer Learning, Model Training & Evaluation, Computer Vision, RAG
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, HuggingFace Transformers, Sentence Transformers, Scikit-learn, OpenCV
- **Backend & API Frameworks:** FastAPI, Flask, Spring Boot, Node.js, Express.js, gRPC
- **Frontend Frameworks:** React, Next.js, TypeScript, Vite, Tailwind CSS
- **Cloud & DevOps:** Docker, Kubernetes, AWS (EC2, VPC, Lambda, S3), Cisco Cloud, CI/CD Pipelines (GitHub Actions), Terraform, Grafana, Prometheus
- **Developer Tools:** Git, GitHub, Jupyter Notebooks, Linux, Agile/Scrum